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Precision-limit positioning of direct drive systems with the existence of friction

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Abstract

This paper discusses the issue of precision-limit positioning of a direct drive system with the existence of friction. By precision-limit positioning it means that, in addition to the common sense of sub-micro, nano, or pico precision, the precision error equals the sensor resolution. This means a zero-count precision error when a digital type position sensor is used in the feedback loop. Precision-limit positioning also enforces 100% repeatability as precision is concerned. Of course, this is the limit of the precision that a real control system can achieve. In the past, this kind of performance was not considered possible mainly for lacking accurate and enough information about friction, especially the static friction. However, as more and more knowledge of the pre-sliding motion has been revealed, it is shown in this research that it is possible to achieve such kind of performance. It also shows that, from the dynamic point of view, to achieve precision limit an integral type control should be included in the final stage of positioning. A simple and explicit criterion is given for the integral control gain to guarantee asymptotic stability. Some practical issues of a digital precision-limit positioning system are also addressed.

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1. Introduction

High precision positioning is an important issue in the fields of manufacturing, assembly and measurement. Recently, the definition of precision has become more and more strict. In this research, we focus on the precision-limit positioning (PLP) of direct drive systems under no or constant load. By PLP it means that, in addition to the common sense of high precision requirement, the precision achieved is the limit that the particular system hardware can achieve. Of course, there are lots of factors that influence the precision limit, such as the resolution of the input force, the resolution of the position sensor, and so on. In this research, the precision limit is defined to be the resolution of the sensor that is used for position feedback. It is under the assumption that all other parts of the system hardware are capable

of achieving the goal of PLP. We further restrict ourselves to the case that a digital type position sensor is used. In this case the precision error should be zero-count error. However, the results and physical phenomena discussed in this paper can also be easily applied to the case when analog sensors are used. It is also noted that to make zero-count-precision-error positioning a meaningful statement, the precision should be 100% repeatable. This is quite different from what is encountered in existing commercial positioning systems. The reason we ask for PLP is two folds. (i) It is just for academic curiosity. We believe that only by pushing the performance to its limit can we see the real physical mechanism in a positioning system. (ii) To cut down the cost of a high precision positioning system by using cheaper sensors, bearings, motors and reducing the associated maintenance cost.

Of course, PLP was not deemed possible in a common sense. This was mainly because we lacked enough knowledge about friction and the induced non-linearity. This ignorance is even worse when we are talking about

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static friction. To overcome the problem caused by friction, the industry uses brushless motors, air bearings, magnetic bearings and other specially designed bearings to get rid of friction. Despite the high costs of these special equipments, the theoretical precision limit for such kind of systems is ± 1 count. This implies 3 times the precision error of ± 0 count case. To get same level of precision, a sensor with triple resolution should be used. This, of course, increases the cost tremendously for a high precision system.

Instead of using expensive hardware, researchers in the control field have been working on complicated control laws to compensate the influence of friction. According to the type of control force that is applied to the system, these designs can be divided into two categories: pulse force and ordinary force. Yang and Tomizuka (1988) might be the first group to propose pulse-width-modulated control. In their design, the control input is a sequence of pulses with constant magnitude while the width of each pulse depends on the position error. Hojjat and Higuchi (1991), on the other hand, proposed amplitude-modulated pulses. In this case the pulse width is fixed. The essential point of these methods is to push the system into the range of dynamic friction by a high force rate. This can avoid the complicated friction behavior in the pre-sliding phase as will be discussed in the next section. However, if the amplitude is not large enough, the pulse cannot break the adhesion of the two contacting surfaces, and it remains stuck. A steady state error then exists due to the work hardening process in this phase (Hsieh & Pan, 2000). Suyama, Izumi and Hitala (1996) and Popovic, Gorinevsky and Goldnberg (2000) follow the same philosophy but use fuzzy reasoning to update magnitude and width. It is noted that, besides the work hardening that may degrade the pulse effect, the pulse force itself contains high frequency signals, which makes it not appropriate for those systems containing flexible structures as discussed in Hsieh and Lin (1997).

As the ordinary control force is concerned, the designs can further be divided into two categories according to the type of friction models adopted for control design. Walrath (1984), Ro and Hubbel (1993) and Feemster, Vedagarbha and Haste (1999) use adaptive control algorithms to compensate the friction non-linearity. The friction model they consider is basically the Amonton–Coulomb type friction, where static friction is considered as a dead-zone non-linearity. The other researches that are based on Amonton–Coulomb type friction include, but not all, genetic algorithm by Huang, Lin and Chen (1998), derivative feedback by Yen, Huang and Lu (1997), combination of fuzzy logic and PD control by Jeon, Kim and Koh (1997), bit control by Otsuka (1992), dither and variable structure control by Guzzella and Glatfelder (1992). All these methods do improve the performance but few of them can achieve

zero-count precision error. On the other hand, some researchers have noticed that the traditional static friction model is not consistent with the real tests; there is a kind of elastic deformation during the stick phase. With this observation, Huang, Yen, and Lu (1999) design the controller based on a dual mode system. In their work, the stick phase is treated as the slip phase but with some uncertainty. A single controller is designed to stabilize both phases. On the other hand, Futami, Furutani and Yoshida (1990), Yamaguchi, Slakey, Saxton, Auslander and Bachand (1990), Iwasaki, Sakai and Matsui (1998), and their coworkers use dual mode integral control. In their design, different control is designed for different phases. Canudas de Wit, Olsson, Aström and Lishinsky (1995) proposed an integrated friction model that covers both static and dynamic friction. However, as static friction, which is critical to PLP, is concerned, this model can only describe part of the hysteresis non-linearity that belongs to the pre-sliding phase. Many important properties, such as congruency of the hysteresis branches, memory and wiping-out effects of the reverse points, creep motion and work hardening phenomena are not addressed (Hsieh & Pan, 2000). An observer-based controller was introduced based on this model.

The very fundamental purposes of these efforts are to improve the precision performance. None of them tried to claim to achieve zero-count precision error with 100% repeatability. However, some works do obtain zero-count precision error in the reports they presented. A common point of these controls is that no matter what kind of algorithms they claim, there is always an integral feedback embedded in the design. The reason for this coincidence will be clear later. Unfortunately, the importance and contribution of this integral control is ignored in all these reports.

In this research, the main purpose is to prove that, with the new knowledge of pre-sliding dynamics revealed, it is possible to obtain PLP rather than just improve the performance. To do this, instead of focusing on complicated control algorithms, we will put most efforts on the dynamics of such kind of systems. Once the dynamic behavior is clear, the conditions under which PLP can be achieved become obvious. Some practical issues on the hardware can also then be addressed. This includes how to select an appropriate D/A converter. The derivation presented in this paper is mainly based on the authors' earlier work on pre-sliding behavior (Hsieh & Pan, 2000). Readers interested in static friction and positioning problems are encouraged to read the paper.

The system used in this research to conduct experiments is shown in Fig. 1. The plant is a direct drive DC torque motor mounted on ordinary ball bearings. The motor is driven by a linear amplifier. Two optical encoders with different resolutions are mounted to

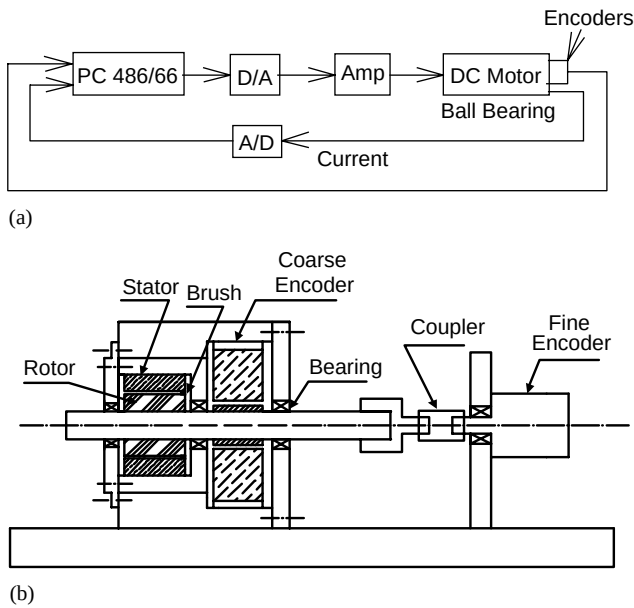


Fig. 1. (a) Block diagram of system hardware, (b) schematic diagram of the test platform.

measure the rotor position. The resolution of the coarse encoder is 86,400 count/rev or, equivalently, 15 arcsec/count, while that of the fine encoder is 1,620,000 count/rev or, equivalently, 0.8 arcsec/count. The rotor angle is measured simultaneously by these two encoders. The outputs are compared to guarantee that the data obtained are correct. However, only one encoder is used in the feedback loop in each test. The data acquisition and control is accomplished by a PC-486/66 together with a D/A converter and an A/D converter. Both D/A and A/D converters have a resolution of 12 bits. The motor parameters such as moment of inertia of the rotor and torque constant are identified using the method developed by Hsieh, Lin, and Pan (1996).

In the next section, the pre-sliding friction behavior will be described. Its influence on the controller design will also be discussed. A simple and explicit stability criterion for fine positioning using PID control is given in Section 3. Theoretical proof for asymptotic stability is briefly described. Some practical issues are also addressed in this section. Section 4 contains the experimental results and is followed by the conclusion.

2. Pre-sliding behavior and its influence on the control design

The traditional friction model is the well-known Amontón–Coulomb friction model. According to this model there is no displacement in the static friction phase. Hence, it is usually modeled as a dead-zone non-linearity in a control design. This assumption is appropriate from the macroscopic point of view.

However, in 1899, Stevens (1899) with the help of an interferometer, found that elastic displacement does exist before the normal slip occurs. This small amount of displacement can indeed be ignored if the precision requirement is not high. However, it becomes the vital factor when we are talking about high precision positioning. Since, in this case, the range of pre-sliding motion is usually greater than the precision requirement by an order of at least 2. The detail of the pre-sliding behavior is described in the work of Hsieh and Pan (2000), and is summarized below.

2.1. Quantitative properties of the pre-sliding behavior

(QN1) *The parameters of any friction model are both time varying and position varying.* It is well known in the field of tribology that it is hard to get repeated friction data. This is true even for well-prepared metal-to-metal friction experiments (Courtney-Pratt & Eisner, 1957), and not to mention what happen in commercial machines. Aging, temperature, humidity, inevitable misalignment of axis, periodically varying motor reluctance and many other factors influence the friction. For example, in a motor system, the maximum static friction at one point may be several times the magnitude of another point. Consequently, any control design under the assumption of fixed friction parameters can hardly achieve PLP. This property is one of the main reasons that people believed that it is impossible to get zero-count precision error with 100% repeatability.

(QN2) *It is temporarily time-invariant and locally position-invariant.* By temporarily time-invariant it means that the parameters remain constant during a short interval of time. This short interval is, however, long enough for ordinary control purposes. By locally position-invariant it means that the parameters remain unchanged if the system stays in the pre-sliding range of any individual point. This is a very important property. Just because of this property, we can discover the following consistent and repeatable qualitative properties. Furthermore, when we make up a model to match these properties, the parameters of this model can be treated as constants with “time-invariant” uncertainties. From the control point of view, it is then possible to obtain zero-count precision error with 100% repeatability. It should be noted, however, this does not mean that we can achieve 100% repeatable transient responses too.

2.2. Qualitative properties of the pre-sliding behavior

The pre-sliding motion consists of two kinds of motion.

(QL1) *Non-linear spring deformation.* This motion demonstrates a special Preisach hysteresis non-linearity,

which has memory and wiping-out effects of reverse points and is congruent in both input and output senses.

(QL2) *Plastic deformation*. This motion demonstrates creep and work hardening.

It is surprising to see that these properties are very close to, if not exactly the same, the structure properties of most plastic materials. It is also noted that these properties are true for the simplest metal-to-metal contact as well as the friction in an ordinary machine where the friction is a combined effect of different friction sources. Remember that the theory of plasticity is restricted to a continuous structure while friction is involved in the interaction of two contacting surfaces. It is still an open question why they behave the same way. Fortunately, Bowden and Tabor (1982) have thrown some light on this subject. They have shown that “it is more accurate to describe surfaces as resembling rolling downs rather than mountain peaks” as is conceived by most people. Furthermore, there is strong adhesion between surfaces. In the pre-sliding phase, friction is mainly the force required to shear these junctions. Consequently, the behavior is dominated by the “average plastic properties of the metal itself near the surface”. This may explain why pre-sliding motion is so close to the structure properties of most mechanical materials.

2.3. Pre-sliding friction model

Based on these properties, a friction model is introduced by Hsieh and Pan (2000) and is shown in Fig. 2(a)¹. This model consists of 4 elements—a plastic module, a non-linear spring module, a viscous damper and a hook. These elements are massless and, hence, do not really exist. They are phenomenological elements. Each element exhibits a special mechanical property of the two contacting surfaces and is described by a simple mathematical expression. At slip phase, the tangential contacting force between the hook and ground can be described as a function of velocity. This kind of function describes the dynamic friction and has been proposed by many investigators. As pre-sliding motion is concerned, the hook sticks at the position where it is, and this model is reduced to that shown in Fig. 2(b). In this model, the damper c is a standard linear damper and the non-linear spring module and plastic module are described below.

Non-linear spring module: Consider the non-linear spring shown in Fig. 3. Let σ_s be the applied external force and x_s be the elongation. The $x_s - \sigma_s$ relation consists of a special Preisach hysteresis as described in property QL1. It has been shown by Hsieh and Lin (2002) that, for such kind of hysteresis that is congruent

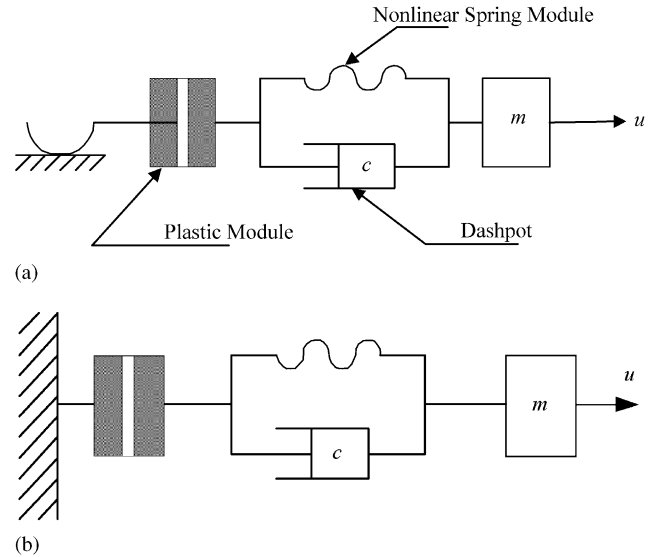


Fig. 2. (a) Model of friction; (b) static friction model.

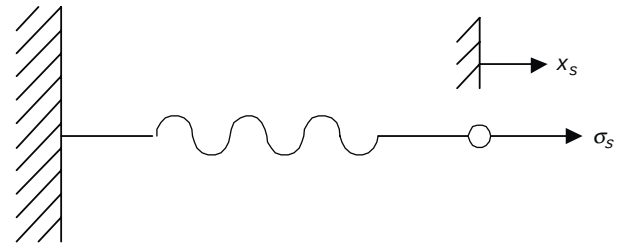


Fig. 3. Non-linear spring module.

in both input and output senses, the constitutive equation of each branch of the hysteretic motion can be expressed as

$$\frac{d\sigma_s}{dx_s} = h(|x_s - x_r|), \quad (1)$$

where $h(\cdot)$ is a non-negative monotonically non-increasing function and x_r is the active reverse point, i.e., the starting point of the current branch of motion. It is noted that the slope at each point is a function of the relative absolute change of x_s with respect to the active reverse point. This is different from the models proposed by Dahl (1976), Canudas de Wit, Olsson, Aström, and Lishinsky (1995) and many other existing models. An appropriate expression of $h(\cdot)$ for pre-sliding motion is

$$\frac{d\sigma_s}{dx_s} = k_1 + k_2 e^{-\beta|x_s - x_r|}, \quad (2)$$

where k_1 , k_2 and β are positive scalars. To get complete information of the hysteretic motion, we have to know the history of x_r . A memory stack can fulfill this purpose. The description of this memory stack is beyond the scope of this paper and is omitted here. A typical response of the non-linear spring is shown in Fig. 4. If we treat the hysteresis as a spring then $h(\cdot)$ can be considered as the spring constant. For very small

¹For convenience, this model is described as a system in linear motion. It should be clear that the analysis and concepts given here are also applicable to rotational motion.

displacement, the non-linear spring can be linearized to be a linear spring.

Plastic module: Consider the plastic module shown in Fig. 5. Let σ be the applied force and x_p be the extension; we have the constitutive equation:

$$\dot{x}_h = \begin{cases} \alpha(f(|\sigma|) - x_h), & \text{if } f(|\sigma|) > x_h, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

$$\dot{x}_p = \text{sgn}(\sigma)\dot{x}_h,$$

where α is a positive scalar and $f(s)$ is a monotonically increasing function with $f(0) = 0$ and $(df(s)/ds)_{s=0} = 0$. In this model, x_h is monotonically non-decreasing and stands for the accumulated work hardening and x_p is the final plastic deformation. It is noted that when $f(|\sigma|) > x_h$, new plastic deformation occurs. This deformation is referred to as the creep motion. On the other hand, if $f(|\sigma|) \leq x_h$, there is no new deformation. The module is work hardened. As a result, $f(|\sigma|)$ stands for the maximum amount of work hardening corresponding to the applied force σ . An appropriate expression of f is $f(|\sigma|) = |\sigma|^n/\lambda$, where $n > 1$ and $\lambda > 0$ are both constants. A typical response of the plastic module is shown in Fig. 6.

The pre-sliding motion is a combined result of these modules according to the relation shown in Fig. 2(b). Hsieh and Pan (2000) have compared lots of experimental and simulation results. They show very good consistency. In a further study, it is shown that the

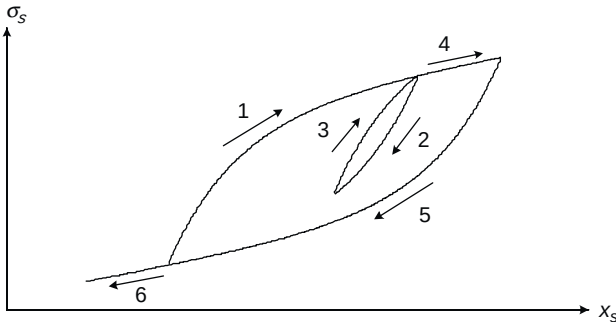


Fig. 4. Typical response of the hysteresis non-linearity of the non-linear spring module.

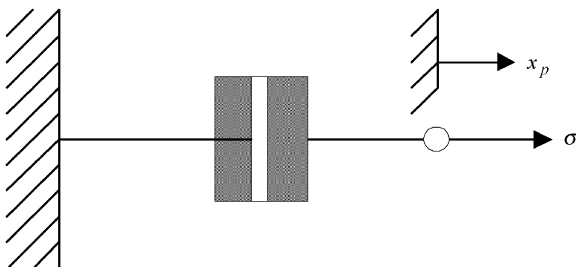


Fig. 5. Plastic module.

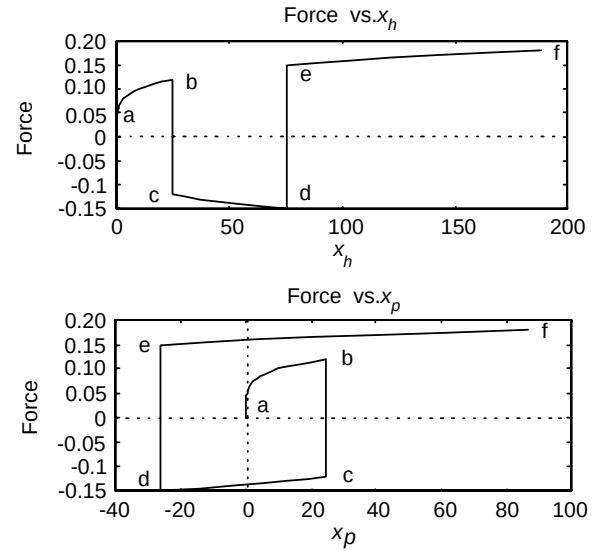


Fig. 6. Typical response of the plastic module.

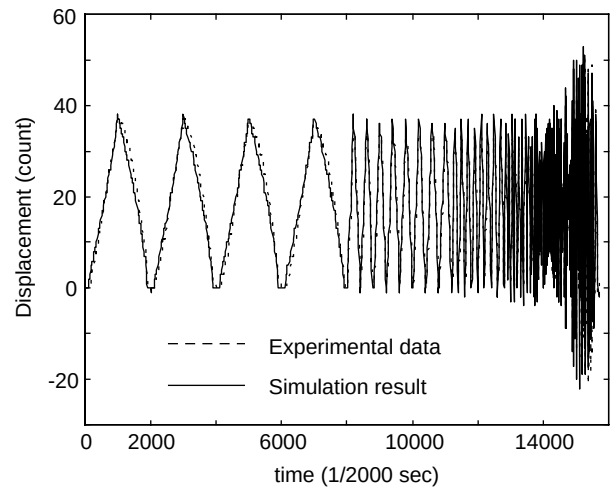


Fig. 7. Dynamic response in the pre-sliding phase. The frequency of input sweeps from 1 to 120 Hz.

parameters of these models are force-rate independent. Fig. 7 shows the simulation and experimental results of the pre-sliding motion of the system shown in Fig. 1. In this experiment the frequency of input signal sweeps from 1 to 120 Hz. Model parameters are fixed in the simulation. The simulation coincides with the experimental data quite well for both high and low input rates.

2.4. Influence of pre-sliding dynamics on control

Let us now examine the position control of a typical direct drive system. The widely accepted governing equation for such kind of system is

$$m\ddot{x} + c_s\dot{x} = u - \tau_f, \quad (4)$$

where τ_f is the friction force, u is the control input and c_s is the damping coefficient in the slip phase. It is noted that, based on the Amonton–Coulomb friction, any point $(x, \dot{x} = 0)$ in the state space is an equilibrium point when u is less than the maximum static friction. Therefore, the control u can be eliminated as the desired position is reached. However, this is not the case in a real system. Careful investigators may have found that if we move a mass block a small amount of distance in the pre-sliding range, it will not stay there when the force is removed. On the contrary, it will bounce back. This is due to the retraction of non-linear spring. This also happens in a positioning system. The control force u becomes small as mass m gets close to the desired position. It finally gets stuck due to static friction. The point it sticks is usually not the desired point. The system behavior switches from that shown in Fig. 2(a) to that in (b). As the plastic module gets work hardened, the behavior of the system is very close to that of a mass-spring-damper system. This is exactly what happens in the steady state. Therefore, to stay at the desired position, say x_d , a final holding force to compensate the spring force is required. If this force is removed, the mass bounces back and leaves the desired position. This fact justifies the necessity of an integral type control in PLP systems, since a non-zero holding force is a typical property of this kind of control. This also explains why in those works (Futami et al., 1990; Yamaguchi et al., 1990; Jeon et al., 1997; Huang et al., 1999; Yen et al., 1997) that show zero-count precision error in their experiments always have an integral part embedded in their feedback loop. This result is quite different from what is claimed by Radcliff and Southward (1990). In their conclusion, an integral control will induce limit cycle. Roughly speaking, this is true when the mass is stuck at a distance beyond the range of pre-sliding motion.

It is clear now that, on the surface, the system given by (4) looks quite simple. It, in fact, consists of two different systems: a type-1 slip system and a type-0 stick system. Since the two systems are so different, it is natural to take different control strategies for different phases to achieve good response. This dual mode control philosophy has been adopted in many research works as mentioned before. The control design for coarse positioning in the slip phase has been well developed. In the next section we will concentrate on the fine positioning in the stick phase.

3. Stability criterion of PID control in the pre-sliding range

3.1. Stability analysis

As discussed in the last section, an integral type control is necessary for a PLP system. Therefore, in this

research we restrict the control structure to be a simple PID control. Though the concept of pre-sliding motion as represented by Fig. 2(b) is simple, the combined dynamics and non-linearity embedded in (1) and (3) still make the mathematical analysis not very tractable. As a first try, we neglect the plastic module in the friction model to simplify the analysis. This is a mild assumption because of two reasons. First, the creep motion is much slower than the non-linear spring deformation. As will be discussed in Section 3.2, the main influence of it is that it needs extra time to keep the system stay still in the range of the desired encoder count. Secondly, as long as the control remains in the pre-sliding phase, the plastic deformation will finally disappear due to work hardening.

With this assumption, the system is thus reduced to that shown in Fig. 8. This is a system with a hysteretic component, and the governing equation for each branch of motion is

$$m\ddot{x} + c\dot{x} + \sigma_s(x) = -k_P e - k_D \dot{e} - k_I \int_0^t e \, d\tau, \quad (5)$$

where $e = x - x_d$, and the relation between σ_s and x follows (1). Differentiating (5) and expressing it in the state space form we have

$$P: \dot{z} = g(z) = \begin{bmatrix} z_2 \\ z_3 \\ -\frac{k_D + c}{m} z_3 - \frac{h(|z_1 + x_d - x_r|) + k_P}{m} z_2 - \frac{k_I}{m} z_1 \end{bmatrix}, \quad (6)$$

where $z \equiv [z_1 \ z_2 \ z_3]^T \equiv [x - x_d \ \dot{x} \ \ddot{x}]^T$. It is noted that the hysteresis non-linearity of the non-linear spring is a system with varying state dimension (Hsieh & Pan, 2000). More specifically, those reverse points x_r 's that have not been wiped out form part of the state of the system. This is quite different from other existing friction models. Standard Liapunov theorem cannot be applied to this system. To analyze the stability of such kind of systems, we treat it as a switched system, a system that switches among an infinite set of subsystems F , $F = \{P| \dot{z} = g(z), x_r \in R\}$. It is obvious that $z = 0$ is the

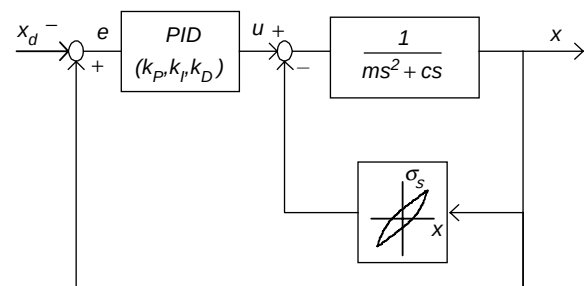


Fig. 8. Control system block diagram.

only equilibrium point for all $P \in F$. This is an autonomous continuous switched system. Since the spring constant $h(\cdot)$ is non-negative and monotonically non-increasing, it is lower and upper bounded by two constants say, $\underline{h}$ and $\bar{h}$. Pan, Lin and Hsieh (2001) have proved that the system in Fig. 8 is globally asymptotically stable about the desired point x_d if the following condition is satisfied:

$$k_I < \underline{k}_I = \frac{(c + k_D)(\underline{h} + k_P)}{m}. \quad (7)$$

To be complete, the main concept of the proof is briefly repeated here. First of all, for each $P \in F$, we can define a Liapunov-like function V_P .

$$V_P = \frac{1}{2m} \begin{bmatrix} z_1 & z_2 \end{bmatrix} \begin{bmatrix} \frac{(c + k_D)k_I}{m} & k_I \\ k_I & h(z_1) + k_P \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \frac{1}{2} \left(\frac{(c + k_D)}{m} z_2 + z_3 \right)^2. \quad (8)$$

If (7) is satisfied, V_P is a positive definite function. It can further be proved that, for all $P \in F$, V_P is upper and lower bounded by two positive definite functions that are independent of P . Differentiate V_P we get

$$\begin{aligned} \dot{V}_P &= \left(k_I - \frac{(c + k_D)(h(z_1) + k_P)}{m} \right) \frac{z_2^2}{m} + \frac{dh(z_1)}{dz_1} \frac{z_2^3}{2m} \\ &= \left(k_I - \frac{(c + k_D)(h(z_1) + k_P)}{m} \right) \frac{z_2^2}{m} + \frac{dh(z_1)}{dt} \frac{z_2^2}{2m}. \end{aligned} \quad (9)$$

It is noted that for each P , when it is active, we have $dh(z_1)/dt \leq 0$. As a result, condition (7) implies $\dot{V}_P \leq 0$, and $\dot{V}_P = 0$ only if $z_2 = 0$. Let $s(\cdot; z_0, t_0, x_{r0})$ be a trajectory starting with $z = z_0$ at $t = t_0$. x_{r0} is the vector, a memory stack, of initial reverse points that are alive at $t = t_0$. $s(\cdot; z_0, t_0, x_{r0})$ is abbreviated as s if there is no ambiguity. It has been explained earlier that even though the system starts with the same z_0 , the response will be different if x_{r0} is different. Let $T(s|P) \equiv \{t_{P0}, t_{P1}, t_{P2}, t_{P3}, \dots\}$ denote the sequence of the end points of the time interval in which subsystem P is active in s , and $I(T(s|P)) \equiv [t_{P0}, t_{P1}] \cup [t_{P2}, t_{P3}] \cup \dots$ be the associated time interval. Let $T(s) \equiv \{t_0, t_1, \dots\}$ be the sequence of the instants that the system switches in trajectory s and $T_e(s) \equiv \{t_0, t_{1-}, t_{1+}, \dots\}$ be the extended sequence of switching instants, where t_{i-} and t_{i+} are the time instants immediately before and after t_i , respectively. Furthermore, let us define the extended Liapunov function V_s of trajectory s , such that $V_s(z) = V_P(z)$ over $I(T(s|P))$. Then it can be shown that $V_s(z)$ is non-increasing at $T_e(s)$. Since $\dot{V}_P \leq 0$ for all active P , we conclude that $V_s(z)$ is non-increasing as $t \geq t_0$. This implies that the system in Fig. 8 is stable in the sense of Liapunov. Furthermore, the norm $\|z\|$ is bounded for each trajectory s . Since $V_s(z)$ is lower bounded and non-increasing, $V_s(z)$ approaches a limit. Let $L([t_i, t_{i+1}])$ be

the length of $[t_i, t_{i+1}]$. Assume $\inf_{t_i \in T} L([t_i, t_{i+1}]) > 0$, then based on some simple concept of real analysis, it can be derived that z_2 approaches 0. From (6), this implies that z approaches 0.

It is noted that result (7) is independent of the exact form of $h(\cdot)$ as long as it is non-negative and monotonically non-increasing. Of course, the globally asymptotic stability claimed here is only for the system described in Fig. 8, which is valid only for the fine positioning phase.

From Liapunov indirect method we can further prove that the system is unstable if

$$k_I > \bar{k}_I = \frac{(c + k_D)(\bar{h} + k_P)}{m}. \quad (10)$$

If $\underline{k}_I < k_I < \bar{k}_I$, the system has a higher and higher possibility to become unstable as k_I approaches $\bar{k}_I$. From the design point of view, condition (7) should be satisfied. This stability condition is exactly the same as that of a standard M–C–K system with K being replaced by $\underline{h}$. It is interesting to note that a pure integral control cannot stabilize system (4). However, it can stabilize the system in the pre-sliding phase as long as $k_I < c\bar{h}/m$.

3.2. Some practical issues of a PLP system

To obtain PLP, the following practical issues should be noticed.

1. $\underline{h}$, same as other friction parameters, is position varying. To get a robust design, the $\underline{h}$ used in (7) must be the lowest non-linear spring constant over the region of working area. Consequently, from property QN2 in Section 2, we know that the PID control satisfying (7) can achieve zero steady state error in all cases and, hence, 100% repeatability of precision is possible. However, this by no means implies repeatable transient response. The values of $\underline{h}$ and $\bar{h}$ can easily be obtained following the process described in Hsieh and Pan (2000).
2. Though $\underline{h}$ is the lower bound of the non-linear spring constant, it is still very large. For example, it is 110 N m/rad for the system used in this research. Therefore, to get a fast response, the integral gain k_I used in the fine positioning phase can be quite large. However, if the same gain is used in the slip phase, it will destabilize the system.
3. The hard non-linear spring also implies that the bandwidth in the pre-sliding phase is much larger than that of the slip phase. For example, the bandwidth of the current system is about 198 Hz in the pre-sliding phase. According to Shannon theorem, the sampling frequency should be at least twice the bandwidth if digital controls are applied. This high sampling frequency is unusual in an ordinary mechanical system.

4. Let Δu be the resolution of the control input. This resolution depends on the bit number of D/A converter, maximum voltage output set on the D/A converter and the gain of the motor driver. It is noted that a change of one unit of the input force will make m to move with an amount of $\Delta u/h(|x - x_r|)$. Suppose Δx is the resolution of the digital sensor used for position feedback then, to get 0-count precision error, we must have $\Delta u/h(|x - x_r|) < \Delta x$. Since $h(\cdot)$ is lower bounded by $\underline{h}$, we must have $\Delta u/h(|x - x_r|) < \Delta u/\underline{h} < \Delta x$ or, equivalently,

$$\Delta u < \underline{h} \Delta x. \quad (11)$$

This equation provides a criterion to choose the D/A converter for a PLP system.

5. Similarly, if the integral feedback is realized using trapezoidal rule

$$u(k) = u(k-1) - k_{ID}(e_{en}(k) + e_{en}(k-1)), \quad (12)$$

where e_{en} is the position error, e , in terms of encoder count, and k_{ID} is the digital integral gain, then, in addition to satisfying the stability requirement (7), we must have

$$k_{ID} < \frac{1}{2} \underline{h} \Delta x. \quad (13)$$

6. When a digital sensor is used in feedback loop, it is not feasible, at the final stage of positioning, to obtain the velocity using difference method, i.e., divide the count difference between two consecutive sampling instants by the sampling period. At the final stage of positioning, the velocity obtained in this way is just a sequence of sparse pulse signals and should not be fed back. The derivative part of the PID control should be avoided in the very final stage. Some commercial machines ignore the velocity signal if it is less than 8 counts/sampling.
7. Since the detailed transition mechanisms between stick and slip is still an open question, the criterion to switch from coarse positioning to fine positioning is still a matter of experience. A conservative way to define the range of fine positioning is the maximum elongation of the non-linear spring. This happens when the maximum static friction τ_s is applied. If the non-linear spring is modeled by (2), then this elongation can be obtained by solving the following inequality (Eq. (14)). But this is just for reference. If creep motion is considered, a much larger range can be assumed:

$$k_1 x + \frac{k_2}{\beta} (1 - e^{-\beta x}) < \tau_s. \quad (14)$$

8. In the above analysis the creep effect is omitted mainly based on the assumption that creep motion is much slower than the non-linear spring deformation. From real-time tests, this assumption is acceptable.

In fact, the main consequence of creep motion is to cause jerks with a magnitude of 1 count in the final stage of positioning. As discussed in the last section, we have to include an integral type control to get zero-count precision error. In most cases, when the desired count is achieved, a final holding force exists to get balance with the non-linear spring force. This holding force will then induce creep motion. The digital sensor cannot detect this creep until it is large enough to get into the range of next count. When this happens, the magnitude of the integral control is decreased by an amount of $2k_{ID}$ in two consecutive sampling instants. The non-linear spring retracts immediately and the sensor shows the desired count again. This process is repeated until the work hardening is completed. According to the friction dynamics described in Section 2, the jerks can happen in only one direction and coincide with the direction of the final holding force. Furthermore, since creep slows down gradually, the frequency of jerks slows down gradually too.

9. It is interesting to note that almost all encoder manufacturers claim ± 1 count precision error for their products. In this case, how could it be possible to get 0 count position error as is observed in all the experiments conducted in this research? As a matter of fact, the meaning of ± 1 count precision error is two folds. First, due to quantization error, a motion of n counts may, in fact, have a displacement that ranges from $(n-1)\Delta x$ to $(n+1)\Delta x$, where Δx is the resolution of the encoder. In other words, there is ± 1 count of uncertainty as displacement is concerned. However, as positioning is concerned, 0 count of position error means that the desired position x_d resides in the range of the specified count. It has an uncertain range that is $\frac{1}{3}$ of those systems that claim ± 1 count position error and $\frac{1}{3}$ of those systems that have ± 2 count error. Secondly, ± 1 count error means that there is a blur region between two adjacent counts. When the system stays in this region, the count number may jump between n and $n+1$, or n and $n-1$. However, for any encoder with acceptable quality, this region is controlled to be very small as compared with Δx . It cannot vibrate among $n-1$, n and $n+1$ if the system stays still. If this happens, according to the working principle of an optical encoder, it cannot even determine the direction of motion correctly. From the discussion in item 8, we know that if the system stays in the blur region of the desired count, the way the count number jumps must coincide with the direction of the final holding force due to creep motion and will be compensated by the integral control. In other words, the integral control will bring the system to the clear region of the desired count finally.

4. Experimental verification

4.1. General description

The experimental system used in this research is described in Section 1. It is a direct drive DC torque motor system equipped with two encoders with different resolutions. The resolution of the fine one is 0.8 arcsec/count (1,620,000 counts/rev), while it is 15 arcsec/count for the coarse one.

First of all, we estimate the bound of the non-linear spring constant, i.e. the lower bound $\underline{h}$ and the upper bound $\bar{h}$. To do this, we follow the process described in Hsieh and Pan (2000) to do static response tests every 30° . This procedure is repeated for 3 circles. The non-linear spring response is then extracted from the data and is shown in Fig. 9. The lower bound $\underline{h}$ is estimated to be $\underline{h} = 110 \text{ N m/rad}$, and the upper bound is $\bar{h} = 860 \text{ N m/rad}$.

As fine encoder is considered, $\underline{h} \Delta x = 4.27 \times 10^{-4} \text{ N m}$. The resolution of input torque of our system is $\Delta u = 2.4 \times 10^{-4} \text{ N m}$. Hence, from (11) we know that this torque resolution is fine enough for sensor-accuracy positioning for both fine and coarse sensors. The moment of inertia m of the rotor is $m = 5.54 \times 10^{-4} \text{ kg m}^2$. The damping coefficient is estimated to be $c = 0.2 \text{ N m/rad s}$. The highest natural frequency of the system is approximately $\omega_n = (\bar{h}/m)^{1/2} = 198 \text{ Hz}$, which occurs when the spring is working around a reverse point. The sampling frequency of our control system is 2 kHz, ten times ω_n .

The experiments include two categories. The first category is to verify the stability criterion as expressed by (7) and (10). The second category is to demonstrate some repeated motions as what are encountered in real applications.

4.2. Category I: Experiments to verify theoretical results

In this category we concentrate on the stability and steady state error of the fine positioning. The coarse positioning and transient response are not the issues. The control strategy is thus a simple dual mode control, where a PI control and a single integral control are used for coarse and fine positioning, respectively. The integral feedback is realized using trapezoidal rule (12). Using bilinear transformation, $k_{ID} \approx k_I T/2$, where k_I is the corresponding continuous feedback gain and T is the sampling interval.

4.2.1. Case 1: With fine encoder feedback

In this case, if we take $\bar{h}$ as the spring constant for control design, we have $\bar{k}_I = c\bar{h}/m = 1.2 \text{ N m/count}$. The corresponding bound $\bar{k}_{ID}$ for the digital system is $\bar{k}_{ID} = 3.01 \times 10^{-4} \text{ N m/count}$. Similarly, if we take $\underline{h}$ as the spring constant for control design, we get $\underline{k}_{ID} = 3.85 \times 10^{-5} \text{ N m/count}$. Three different gains are tested for stability.

- $k_{ID} = 3.85 \times 10^{-5} \text{ N m/count}$. In this case, $k_{ID} = \underline{k}_{ID} < \frac{1}{2} \underline{h} \Delta x$. Both (7) and (13) are satisfied. The system is stable in all tests as predicted by the theory. Fig. 10 shows the results of 10 successive experiments with the desired position starting from 1 count to 10 counts. They all achieve zero-count error. Fig. 11 shows the results when a desired position of 30 counts is commanded. The jerks created by creep motion appear. The direction of jerks coincides with the direction of final holding force. It is clear that the frequency of jerks slows down gradually.
- $k_{ID} = 9.45 \times 10^{-5} \text{ N m/count}$. In this case, $\underline{k}_{ID} < k_{ID} < \bar{k}_{ID}$. It is stable in some cases, but is

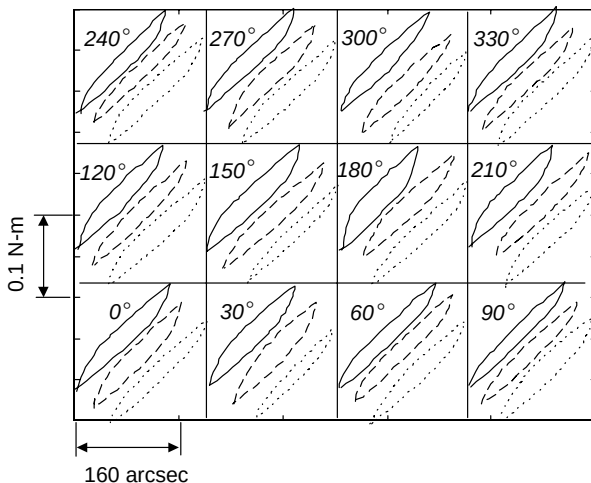


Fig. 9. Response of the non-linear spring at different positions.

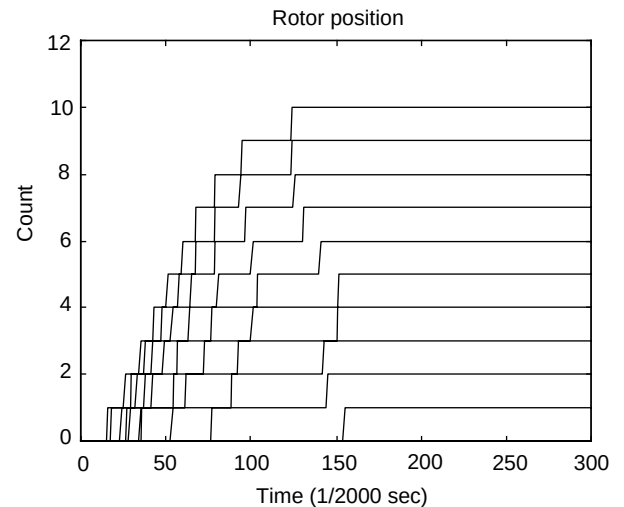


Fig. 10. Fine positioning with $k_{ID} = 3.85 \times 10^{-5} \text{ N m/count}$.

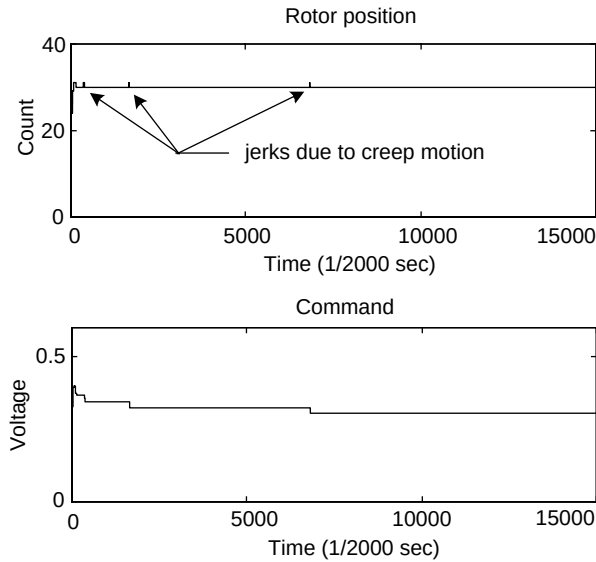


Fig. 11. Fine positioning—jerks appear due to creep motion.

unstable in most cases. To stabilize the system, we add a proportional control with $k_P = 6.21 \times 10^{-4} \text{ N m/count}$. In this case, stability criterion (7) is satisfied again. The system becomes stable in all the cases we tested. Fig. 12 shows the response of 10 successive experiments. The responses are much faster as compared with those in Fig. 10. But overshoot occurs.

- (c) $k_{ID} = 2.08 \times 10^{-4} \text{ N m/count}$, which is about 70% of $\overline{k_{ID}}$. It only remains stable once in all our tests. When $k_{ID} > \overline{k_{ID}}$ the system is unstable in all tests.

In the above tests, the whole process stays in the pre-sliding phase. In what follows, we do long distance tests with coarse sensor being used in the feedback loop.

4.2.2. Case 2: With coarse encoder feedback

In this case, $k_{ID} = 7.22 \times 10^{-4} \text{ N m/count}$, and $\overline{k_{ID}} = 5.64 \times 10^{-3} \text{ N m/count}$. The integral gain in the fine positioning phase is $k_{ID} = 7.22 \times 10^{-4} \text{ N m/count}$, which equals $\overline{k_{ID}}$. The integral gain in the coarse positioning is $7.22 \times 10^{-6} \text{ N m/count}$, $\frac{1}{100}$ of the fine positioning gain. The proportional gain in the coarse positioning phase is $2.91 \times 10^{-3} \text{ N m/count}$. The response of a maneuver of 21,600 counts (90°) is shown in Fig. 13. Zero-count error is achieved. These experimental results completely match the theoretical results.

4.3. Category II: Positioning of repeated motions

In these experiments, PD control and PID control are adopted for coarse and fine positioning, respectively. But the control gains vary according to the level

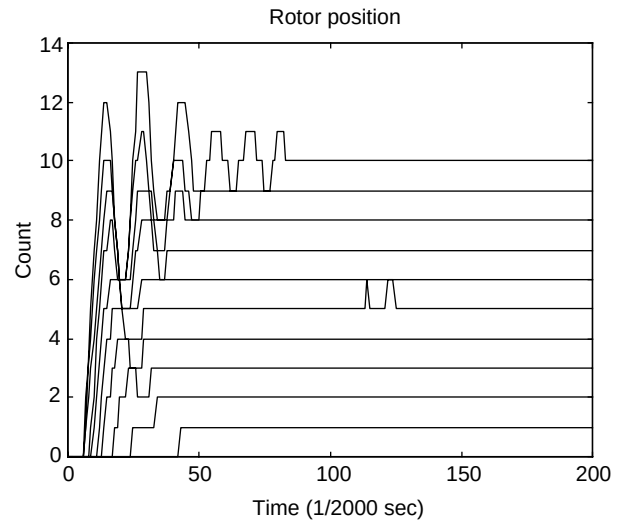


Fig. 12. Fine positioning with $k_{ID} = 9.45 \times 10^{-5} \text{ N m/count}$ and $k_P = 6.21 \times 10^{-4} \text{ N m/count}$.

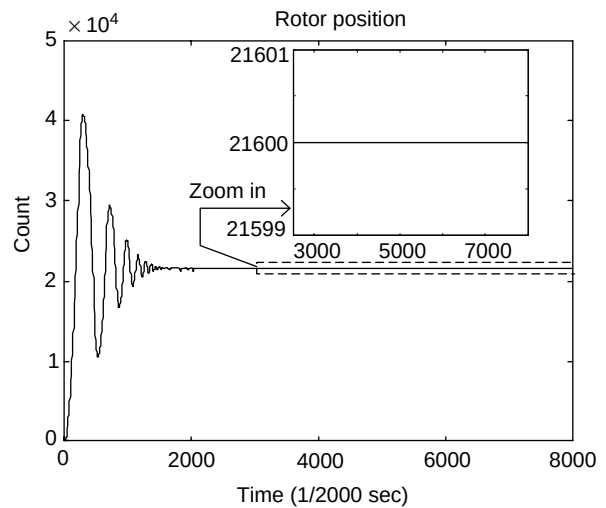


Fig. 13. Long-distance positioning with desired position being 21,600 count.

of position error to avoid overshoot. Since it is not the main issue of this paper, the detail of control strategy is omitted.

The switching condition between coarse and fine positioning is as follows. The control switches from coarse to fine positioning if (i) $e_{en} \leq 150$ counts of coarse encoder and (ii) $\sum \Delta e_{en} \leq 1$ for 20 consecutive sampling times, where Δe_{en} is the difference of e_{en} between two consecutive sampling instants. The control switches from fine to coarse positioning whenever $e_{en} > 150$ counts.

Fig. 14 shows the responses to square wave commands and Fig. 15 the responses to staircase commands. The steady state error in each case is 0 count.

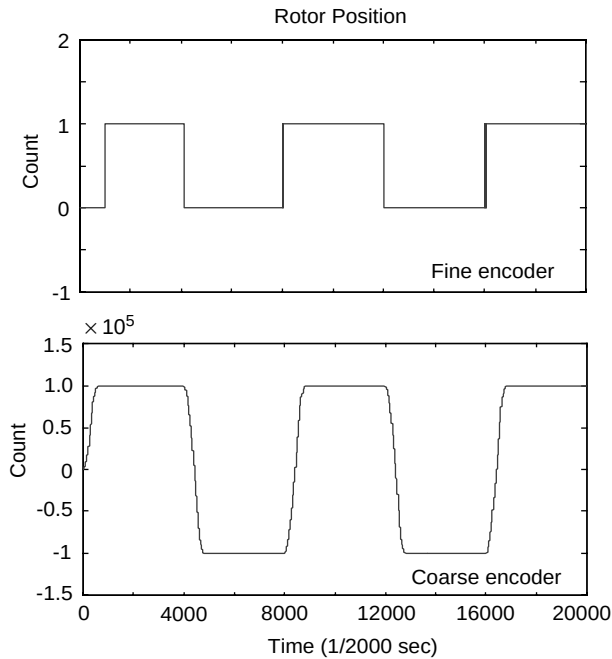


Fig. 14. Responses to square wave commands.

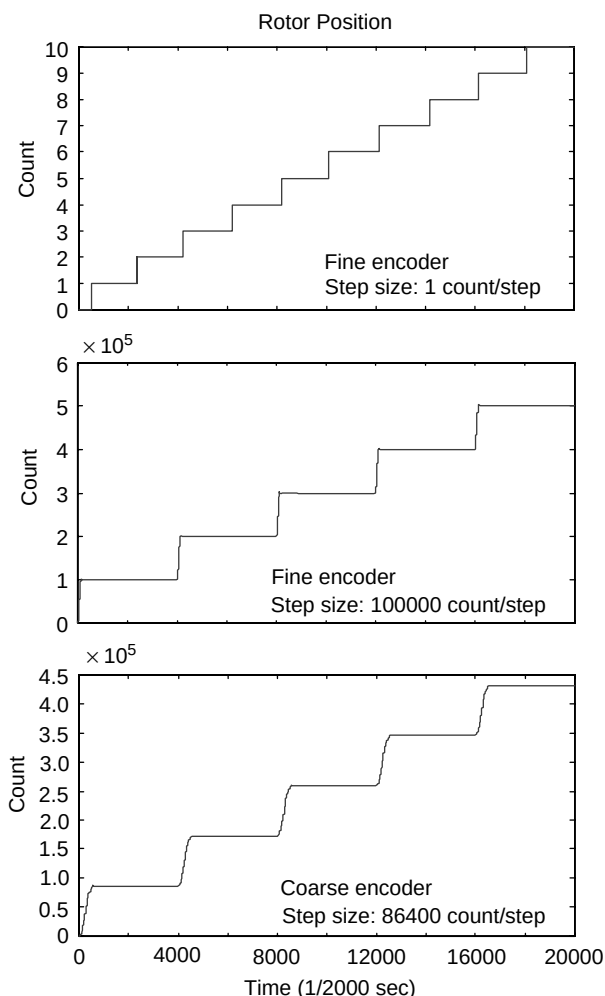


Fig. 15. Responses to staircase commands.

5. Conclusions

In this paper, it has been shown that it is possible to achieve precision-limit positioning for direct drive systems. As precision is concerned, this kind of performance is the limit a real positioning system can achieve. This achievement is mainly based on the new knowledge of the pre-sliding dynamics revealed recently. It turns out that to get precision-limit positioning (PLP) an integral type control should be included in the closed loop. A simple and explicit stability criterion is given. The results have been tested for the past several years, and the system responses coincide with what are predicted by theoretical analyses. The current issue of interest is to eliminate the jerk induced by the creep motion at the final stage of positioning.

Acknowledgements

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